

RESTLESS LEG SYNDROME

**MUHAMMAD SYAHIRUL KHALIQ BIN MOHAMED AIDI
SHAHRIZ**

**FACULTY OF COMPUTER SCIENCE AND INFORMATION
TECHNOLOGY
UNIVERSITI MALAYA
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RESTLESS LEG SYNDROME

MUHAMMAD SYAHIRUL KHALIQ BIN MOHAMED AIDI SHAHRIZ

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TECHNOLOGY
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Name of Candidate: Muhammad Syahirul Khaliq bin Mohamed Aidi Shahriz
(I.C/Passport No: 980910-56-5139)

Matric No: 22075208

Name of Degree: Master of Data Science

Title of Project Paper/Research Report/Dissertation/Thesis ("Restless Leg Syndrome"):

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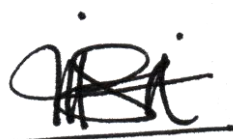
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Name: Ts. Dr. Mohd Shahrul Nizam Mohd Danuri

Designation: Senior Lecturer

Abstract

This study focuses on classifying restless leg syndrome (RLS) in aid of anxiety detection in different activities via machine learning techniques. The collection of data involves capturing leg movements of 3 RLS and non-RLS participants using a mobile phone app called Advanced Physical Toolbox based on 3 separate activities, which are sitting still, watching horror movies and playing piano. The data collected will primarily focus on z-axis movement in relation to previous study of RLS diagnosis. The raw data undergoes preprocessing techniques such as low pass filtering and Fast-Fourier Transformation process to reduce noise and obtain data in both time and frequency domain. Subsequently, statistical, motion and Fourier-transformed based feature were extracted and selected via Principal Component Analysis (PCA) and feature importance-based selection via Random Forest algorithm. K-means clustering was performed to divide clusters into 2 labels, namely RLS and non-RLS. Naïve Bayes (NB), K-Nearest Neighbor (KNN), Support Vector Machine (SVM) and Multilayer Perceptron (MLP) models were developed and assessed with their ability to classify RLS across different activities. The result indicates MLP outperforms other models in terms accuracy, precision, recall and F1-score for each activity. While SVM performs relatively well in general, NB and KNN exhibits poor performance with significant drop of performance scores and underfitting issue in 2 activities. In summary, advanced models are preferred due to complex patterns and non-linear relationships of classification process.

Abstrak

Kajian ini memberi tumpuan kepada mengklasifikasikan ‘Restless Leg Syndrome’ (RLS) dalam membantu pengesanan ‘anxiety’ dalam aktiviti yang berbeza melalui teknik pembelajaran mesin. Pengumpulan data melibatkan menangkap pergerakan kaki 3 peserta RLS dan bukan RLS menggunakan aplikasi telefon mudah alih yang dipanggil ‘Advanced Physical Toolbox’ berdasarkan 3 aktiviti berasingan, iaitu duduk diam, menonton filem seram dan bermain piano. Data yang dikumpul akan tertumpu terutamanya pada pergerakan paksi-z berhubung dengan kajian terdahulu tentang diagnosis RLS. Data mentah menjalani teknik prapemprosesan seperti penapisan laluan rendah dan proses Transformasi Fast-Fourier untuk mengurangkan hingar dan mendapatkan data dalam domain masa dan frekuensi. Selepas itu, ciri berasaskan statistik, gerakan dan Fourier-transformed telah diekstrak dan dipilih melalui Analisis Komponen Utama (PCA) dan pemilihan berdasarkan kepentingan ciri melalui algoritma Random Forest. K-means clustering dilakukan untuk membahagikan cluster kepada 2 label iaitu RLS dan bukan RLS. Model Naïve Bayes (NB), K-Nearest Neighbor (KNN), Mesin Vektor Sokongan (SVM) dan Multilayer Perceptron (MLP) telah dibangunkan dan dinilai dengan keupayaan mereka untuk mengklasifikasikan RLS merentas aktiviti yang berbeza. Hasilnya menunjukkan MLP mengatasi model lain dari segi ketepatan, ketepatan, ingatan semula dan skor F1 untuk setiap aktiviti. Walaupun SVM menunjukkan prestasi yang agak baik secara umum, NB dan KNN mempamerkan prestasi yang lemah dengan penurunan markah prestasi yang ketara dan isu ketidaksesuaian dalam 2 aktiviti. Ringkasnya, model lanjutan lebih disukai kerana corak kompleks dan hubungan bukan linear proses pengelasan.

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List of Symbols and Abbreviation

For examples:

WED	:	Willis-Ekbom disease
RLS	:	Restless Leg Syndrome
CBT	:	Cognitive Behavioural Therapy
EMG	:	Electromyography
PSG	:	Polysomnography
EDA	:	Exploratory Data Analysis
PCA	:	Principal Component Analysis
RUS	:	Random Undersampling
SMOTE	:	Synthetic Minority Over-sampling Technique
EDA	:	Exploratory Data Analysis
FFT	:	Fast Fourier Transform
NB	:	Naïve Bayes
SVM	:	Support Vector Machine
KNN	:	K-Nearest Neighbors
MLP	:	Multilayer Perceptron

Chapter 1: Introduction

1.1 Research Background

Anxiety can be defined as a combination of fear and future-oriented mood state associated with preparation of potential unpleasant and bad events (Barlow, 2004). Anxiety disorder is one of the most significant mental health issues nowadays. Statistics has shown that 4.05% of worldwide population suffers from anxiety disorder with number of cases increases by 55% for the past 3 decades (Javaid et al., 2023). The aftermath of pandemic has also resulted in increase of anxiety disorders, particularly among graduate students (Chi et al., 2023). Such issue significantly impacts individual's life quality, productivity, and overall well-being. A recent study indicated that a person diagnosed with anxiety disorder does not only exhibit problems of their working performance, but also encounters issues of their daily occupation, which includes household management, socialization and personal care (Broström et al., 2024). Thus, early detection may prove beneficial in prevention of subjected condition to worsen.

While traditional methods can be subjective and time-consuming, the advancement of technology in detecting physiological indicators of anxiety provides a more favourable alternative. A study was conducted on monitorization and detection of anxiety via Zephyr HxM Bluetooth band and highlighted the success of heart rate as a consistent and reliable indicator of social anxiety disorder detection (Miranda et al., 2014). Another study conducted on identifying anxiety via Shimmer2 unit (Shimmer Sensing Ltd.) to obtain the data of respiratory signals and electrocardiogram (ECG) among autistic patients, in which results indicate high relation of deep breathing initiation to anxiety cases (Nguyen et al., 2021). These findings highlight the potential of technological devices in indicating anxiety spans and continuous tracking of it, providing a solution that can overcome barriers from traditional methods.

In recent researches, Restless Leg Syndrome can be significantly associated with anxiety disorder. Also known as *Willis-Ekbom disease* (WED), restless leg syndrome is defined as the urgency to move the legs because of uncomfortableness which can only be slightly or fully eased with movements. Such disorder can occur mainly during evening or night time (Walters et al., 1995). A systematic review was conducted via databases involving researches since 2023, highlighting the presence of anxiety in RLS patients are significantly higher compared to non-diagnosed patients (An et al., 2024). Despite not having a direct causation between RLS and anxiety, the significant correlation between both can be used as a tool to predict and detect anxiety.

1.2 Research Problem Statement

While anxiety detection has been a research interest these days, there are several limitations in existing research that requires further exploration. A comprehensive review of anxiety studies have indicated the lack of non-invasive sensors utilization to provide physiological signals for anxiety detection as well as non-conclusive results of available studies (Jiménez-Mijangos et al., 2023). Another study has indicated its limitation from the absence of individual's daily physical activities input such as accelerometer data in predicting anxiety via technological tool (Nath & Thapliyal, 2021). This is essential to reduce biasness from self-reported measures and increase accuracy in anxiety prediction.

Machine learning studies has been growing in the sector of restless leg syndrome prediction via various methods. A recent research applied a convolutional neural network classifier to differentiate RLS and normal patients based on their cortical activities (M. Kim et al., 2022). However, machine learning studies on RLS accelerometry for diagnosis are still relatively limited. Research addresses its limitation on accelerometry data analysis using

Hidden Semi Markov Models, emphasizing the need of machine learning to provide better accuracy and model optimization (Van Kuppevelt et al., 2019). Another research have also suggested the requirement of leg accelerometry to validate the RLS diagnosis and severity, which are currently lacking in its study (Nath & Thapliyal, 2021). Therefore, these studies signify the clear need of integrating machine learning in accelerometry studies for RLS prediction.

Therefore, this research aims to address these gaps by evaluating the efficacy of various machine learning algorithms for predicting anxiety specifically via leg accelerometry.

1.3 Research Questions

1. How can accelerometer data captured from leg movements be utilized to differentiate between states of anxiety and non-anxiety based on restless leg syndrome?
2. How can machine learning techniques aid in detection of anxiety through leg movements data captured?
3. Which machine learning algorithms perform best in classifying anxiety based on accelerometer data?

1.4 Research Objectives

1. To collect and preprocess accelerometer data capturing leg movements in individuals with and without restless leg syndrome (RLS).
2. To develop machine learning models based on classifying accelerometer data into RLS and non-RLS.
3. To evaluate the performance of different models using metrics such as accuracy, precision, recall, and F1-score

1.5 Research Scope and Contribution

The focus of this research involves utilization of accelerometer data in capturing the trends and patterns of restless leg syndrome (RLS) as well as classifying for RLS diagnosis to enhance the detection of anxiety. The scope of study includes several key aspects, including collection of data via capturing leg movements in individuals while carrying out several respective activities. Data capturing process includes utilizing android smartphone with an app called Advanced Physical Toolbox, which presents the data in a structured excel format. This allows for easy data manipulation and analysis, particularly in real time cases. The study includes preprocessing techniques, feature extraction and target labelling from clustering from raw data collected to gain insights and patterns for machine learning model development. This study also includes utilization of several proposed machine learning techniques from past researches to obtain the prime model in classification process. The model will be evaluated based on suggested evaluation metrics across 3 different activities in separate manner to focus on accuracy and generalizability of model.

The contribution of this research includes improving the accuracy and objectivity of anxiety detection, potentially reducing reliance on subjective self-reports. Integrating machine learning techniques in classifying accelerometer data for restless leg syndrome (RLS) can lead to more precise and reliable diagnostic tools. Furthermore, this research can lead to the development of wearable devices that utilize accelerometer data to detect anxiety and RLS in real-time, providing healthcare providers with accurate, non-invasive diagnostic tools. This could revolutionize patient monitoring, leading to earlier interventions and better management of these conditions. The findings can be applied to enhance the functionality of smart devices like fitness trackers and smartwatches, allowing them to monitor users' mental health and sleep disorders, offering personalized health recommendations. In addition, integrating these tools

into corporate wellness programs, companies can proactively address mental health issues, leading to a healthier workforce.

Chapter 2: Literature Review

2.1 Anxiety disorder

Anxiety disorder has a long history, where similar symptoms of it were diagnosed in cultural text but never recognized as a mental health problem. Anxiety was particularly associated with spiritual or moral flaws, with religious or mystical involvements focused as the method of treatment (Kelly, 2002). The term "anxiety disorder" became formally acknowledged through the development of diagnostic criteria and the advent of cognitive-behavioural therapies (CBT) as successful treatments in the mid-20th century. Anxiety disorders often manifest early in life, with a significant portion of cases emerging during childhood or adolescence (Kessler et al., 2005). The root cause involves genetic predisposition, as well as environmental stressors including trauma, chronic stress, and negative childhood incidents, that has the ability to trigger or worsen symptoms (Smoller et al., 2009). The Diagnostic and Statistical Manual of Mental Disorders (DSM) has since refined different types of anxiety disorders such as generalized anxiety disorder, panic disorder, and social anxiety disorder, in such that these disorders are generally related but have unique conditions as well (Joyce-Beaulieu & Sulkowski, 2016).

Traditional methods for diagnosing and treating anxiety disorders are deemed to be subjective and potentially bias as it completely depends on interviews and self-report questionnaires. Some of the customary treatment approach including cognitive-behavioural therapy (CBT) and pharmacotherapy are not particularly effective for all with even some may exhibit residual symptoms post treatment (Cuijpers et al., 2014). As a result, there has been a deep exploration in implementing technology such as smartwatches and mobile apps with the addition of machine learning algorithms to aid the diagnosis of anxiety disorders as it allows real-time monitoring and a more specific method of interventions to be manifested. For

instance, studies have shown potential development of non-invasive diagnostic tool in detection of anxiety states via physiological data of heart rate variability (HRV) and skin conductance (H.-G. Kim et al., 2018).

However, there are several limitations in the current research on anxiety detection and treatment using technological tools. One major challenge is the lack of integration between physiological data and daily activity patterns, which can provide a more comprehensive understanding of an individual's mental state. Moreover, most studies have focused on short-term or situational anxiety rather than chronic anxiety disorders, limiting the generalizability of their findings. There is also a need for larger, more diverse samples to validate the effectiveness of these technologies across different populations. Addressing these gaps could lead to more accurate and personalized approaches to managing anxiety, ultimately improving outcomes for those affected by these pervasive disorders (Jiménez-Mijangos et al., 2023; Nath & Thapliyal, 2021)

2.2 Non-invasive sensors on behavioural detection

Utilization of non-invasive sensors have become a rising trend recently due to unrequired intrusive procedures to detect and keep track of physiological and behavioural data. These sensors provide valuable needs particularly in healthcare industry which require continuous assessment to monitor chronic conditions or detect minor changes in patient's health. Examples of non-invasive sensors are wearables with tracking heart rate, blood pressure, and glucose levels ability, as well as environmental sensors that monitor air quality and temperature (Patel et al., 2012). The growing interest in such devices stems from their ability to deliver real-time data, subsequently improving patients' diagnosis as well as cutting down requirements of invasive diagnostic procedures. In addition, the development of sensor

technology and the components reduction in size creating devices with better comfortability, more user-friendly, and greater accessibility to a broader population.

One of the most common usages of non-invasive sensors is remote health monitoring, particularly for treatment of chronic diseases including diabetes, hypertension, and cardiovascular conditions. Wearable sensors such as smartwatches and fitness trackers enable early interventions and personalized treatment via continuous monitoring of vital signs which provide valuable information on a patient's health status. For example, interstitial glucose levels can be measured via non-invasive sensors called continuous glucose monitors (CGMs), producing real-time data on blood sugar levels of diabetic patient while eliminating the need for finger-pricking (Bergental et al., 2018). Similarly, wearable ECG monitors can detect irregularity of heart rhythms in alerting patients or healthcare providers on potential cardiac events, subsequently improving supervision of conditions such as atrial fibrillation (Ghosh & Mishra, 2019). These technologies increasing potential shift from reactive to proactive approach in healthcare industries, thus highlighting prevention and timely detection.

Furthermore, non-invasive sensors are highly explored for their function in mental health management, particularly in detecting and monitoring anxiety, stress, and depression. A previous study indicated the utilization of non-invasive sensors in stress levels and emotional state assessment via physiological signals including heart rate variability (HRV), skin conductance, and electrodermal activity (EDA) (Ming-Zher Poh et al., 2010). These signals produce complimentary data with traditional self-reported measures to eliminate subjectivity and biasness. Furthermore, the incorporation of machine learning algorithm in sensors data further allows development of predictive models for pattern identification associated with mental health conditions and trigger timely interventions. Although, challenges especially in establishing accuracy and reliability of the data captured by these sensors persists. As research in this area continues, refining non-invasive sensor technologies and integration into clinical

practice plays an essential role for global adoption and effectiveness guaranteed in healthcare industries.

2.3 Restless Leg Syndrome (RLS)

Restless Leg Syndrome (RLS) is a neurological disorder characterized by an irresistible urge to move the legs, often accompanied by uncomfortable sensations. These symptoms typically worsen during periods of inactivity, such as when lying down or sitting, and are relieved by movement. RLS is a chronic condition that can significantly impair sleep quality and overall quality of life. The pathophysiology of RLS is not fully understood, but it is believed to involve dysfunctions in the dopaminergic system and iron metabolism within the central nervous system (CNS) (Allen et al., 2014). Genetic predisposition also plays a role, as evidenced by the identification of several genetic loci associated with RLS in genome-wide association studies (GWAS) (Schormair et al., 2017). Despite these insights, the exact mechanisms underlying RLS remain elusive, highlighting the need for continued research into its causes and treatment.

Traditionally, the diagnosis of RLS has relied on clinical criteria and patient self-reports, which can be subjective and prone to misinterpretation. The International Restless Legs Syndrome Study Group (IRLSSG) has established diagnostic criteria that emphasize the subjective nature of symptoms, such as the urge to move and symptom relief through movement (Allen et al., 2003). However, the variability in symptom presentation and the overlap with other conditions, such as peripheral neuropathy, can complicate the diagnostic process. Consequently, there has been growing interest in objective measures to support the diagnosis of RLS. Polysomnography and actigraphy have been used to assess periodic limb

movements during sleep, which are common in RLS patients, but these methods are not always practical for widespread clinical use due to their cost and complexity (Michaud et al., 2002).

Current advancement in wearable technology and machine learning provides new approach on evaluation of RLS. Accelerometers turns out to be one of the non-invasive techniques suggested in detection of leg movement traits corresponding with RLS. Previous study have indicated the identification and quantification of periodic limb movements through accelerometer data, thus allowing it to become a potential tool for RLS diagnosis and monitoring (Van Kuppevelt et al., 2019). However, the application of machine learning to this data is still in its early stages. Most existing studies have focused on basic movement detection algorithms, with limited exploration of more sophisticated techniques like deep learning. Furthermore, there is a need for large-scale studies to validate these approaches and establish standardized protocols for their use in clinical practice (M. Kim et al., 2022). Integrating machine learning with accelerometry could significantly enhance the accuracy and reliability of RLS diagnosis and severity assessment, addressing some of the current limitations in the field.

2.4 Accelerometer

Accelerometers are motion-sensing devices that measure the rate of change in velocity, providing data on movement dynamics and orientation. Originally developed for industrial applications, accelerometers have become widely used in various fields, including healthcare, sports, and consumer electronics. These sensors work based on the principle of detecting changes in acceleration along one or more axes, typically using microelectromechanical systems (MEMS) technology. This allows for the precise measurement of movement patterns in three dimensions (x, y, and z axes), making accelerometers ideal for applications requiring

detailed motion analysis (Woodman, 2007). The versatility of accelerometers is enhanced by their ability to be embedded in portable devices such as smartphones, fitness trackers, and wearable health monitors, enabling continuous, real-time data collection in a variety of settings.

In the healthcare domain, accelerometers have been extensively used for monitoring physical activity, assessing gait, and detecting falls, particularly in elderly populations. These sensors provide objective data that can be used to track the intensity, duration, and frequency of movements, offering insights into a person's activity levels and overall physical health (Godfrey et al., 2008). For example, accelerometers are commonly used in studies to quantify daily physical activity in patients with chronic diseases like Parkinson's disease, where they can help monitor disease progression and the effectiveness of interventions (Müller et al., 2017). Additionally, accelerometers play a crucial role in rehabilitation, where they are used to track patient adherence to exercise regimens and evaluate recovery after surgery or injury. The ability to capture fine-grained movement data has made accelerometers indispensable in both clinical and research settings.

Beyond physical health, accelerometers are increasingly being utilized in mental health research, particularly in the study of conditions such as anxiety, depression, and Restless Leg Syndrome (RLS). By capturing patterns of movement, accelerometers can provide insights into behaviours associated with these conditions, such as restlessness or agitation. For example, in anxiety research, accelerometer data can be used to objectively measure physical manifestations of anxiety, such as fidgeting or pacing, which are often difficult to quantify through self-reports alone (Teo et al., 2019). In the context of RLS, accelerometers have been employed to monitor leg movements during sleep, providing a non-invasive means of diagnosing and assessing the severity of the condition (Van Kuppevelt et al., 2019). The integration of accelerometer data with machine learning algorithms has further enhanced the ability to analyse complex movement patterns, opening new avenues for the early detection

and personalized management of various health conditions. Despite their wide-ranging applications, challenges remain in ensuring the accuracy, calibration, and interpretation of accelerometer data, particularly in complex or uncontrolled environments.

2.5 Machine Learning Approach in accelerometer-based activity detection

Studies on accelerometer-based activity via machine learning have been a hot topic in the research industry recently. Table 1 illustrates the related studies on anxiety detection via machine learning.

Table 1: Critical Analysis of accelerometer-based activity study via machine learning

Title	Dataset	Highlighted Algorithm/Techniques	Result	Limitation	Reference
Machine Learning-Based Anxiety Detection in Older Adults Using Wristband Sensors and Context Feature	Physiological signals (electrodermal activity (EDA) and blood volume pulse (BVP)) with a context-based feature	Random Forest, Logistic Regression and Support Vector Machine (SVM) for classification of anxiety and non-anxiety	EDA and BVP accuracy are 89% and 78% respectively. Addition of context feature improves the accuracy of the models by 3.37% and 6.41 % respectively	• Small sample size limits generalizability • Focuses on healthy participants and does not include individuals with clinical anxiety disorder • Complexity of combining different physiological signals in real time using low cost device	(Nath & Thapliyal, 2021)
Design of a deep learning model for automatic scoring	800 overnight polysomnography (PSG)	Deep Neural Network	Current model outperform 5 existing periodic leg movement detectors in	• Data and validation test used from specific cohorts, which may limit the generalizability	(Carvelli et al., 2020)

of periodic and non-periodic leg movements during sleep validated against multiple human experts	recordings via electromyography (EMG) channel		terms of F1-score, as well as exhibit a better precision and recall in 4 out of 5 models	Reliance from EMG data may overlook other factors, affecting the accuracy for different context	
Validation of an algorithm for the diagnosis of Restless Legs Syndrome: The Restless Legs Syndrome-Diagnostic Index (RLS-DI)	Structured telephone interviews and sleep lab records	A new diagnostic algorithm called Restless-Leg Syndrome Diagnostic Index (RLS-DI)	The RLS-DI showed high diagnostic accuracy with a sensitivity of 93.0% and specificity of 98.9%. It successfully diagnosed 96.1% of the patients, in other words able to separate RLS with other sleep related disorders	- Reliance on a single sleep lab limits generalizability - Requires diverse population in different clinical settings for further improvement - Questionnaire from interviews may result in biasness	(Beneš & Kohnen, 2009)

Detection and Classification of anxiety in university students through the application of machine learning	Physiological signals, such as heart rate, skin conductance, and movement, as well as questionnaire for self-reported anxiety level	Naïve Bayes, Decision Tree, Support Vector Machine (SVM), and Random Forest	Accuracy for naïve bayes, decision tree, random forest and SVM are 71.05%, 71.05%, 78.9% and 75.5% respectively.	• Small sample size of students limits generalizability • Highly variable data, which can affect models' performance • Self-reported questionnaire may introduce biasness	(Bhatnagar et al., 2023)
Sleep syndromes onset detection based on automatic sleep staging algorithm	30 seconds long of epoch Electroencephalogram (EEG) recordings via FFT	Deep Convolutional LSTM Neural Network	Accuracy, precision, recall and F1-score of 90.43%, 77.76%, 93.32%, and 89.12% respectively with mean false error loss of 0.09	• Insufficient data of less distinct sleep patterns impacted on generalizability of the model	(Cvetko & Robek, 2021)

Explainable Machine-Learning-Based Characterization of Abnormal Cortical Activities for Working Memory of Restless Legs Syndrome Patients	EEG recordings from both RLS patients and healthy control subjects	Convolutional Neural Network (CNN) Classifier	Accuracy of 99.32% and 94.5% for training and test. True positive and negative are 95.67% and 91.27% respectively.	• CNN has high complexity and designing such model requires cross-domain knowledge and time-consuming	(M. Kim et al., 2022)
Accelerometer-measured physical activity	Accelerometer of physical activity and sleep quality of	-	-	• Low variance of physical activity does not indicate any proper conclusion on study • Does not involve prediction	

and its impact on sleep quality in patients suffering from restless legs syndrome	patients collected through actigraph links				
Physical Activity Monitoring and Classification Using Machine Learning Techniques	Physical activities recorded using Samsung Galaxy S2 3D gyroscope and accelerometer sensors	Utilized Fast Fourier Transform (FFT) to extract trend from frequency domain	Support Vector Machine achieved highest F1-score between 0.8032 to 0.9638	• Dataset collected from only healthy people, thus limits generalizability • Occurrence of data imbalance affects performance scores	(Alsareii et al., 2022)

A machine-learning approach for stress detection using wearable sensors in free-living environments	SWEET dataset collected from 240 subjects via electrocardiography (ECG), skin temperature (ST), and skin conductance (SC)	K-Nearest Neighbor (KNN), Support Vector Classification (SVC), Decision Tree (DT) and Random Forest (RF)	K-Nearest Neighbor performs best with 98% accuracy	Data imbalance issue are not fully solved	(Abd Al-Alim et al., 2024)
Activity recognition using accelerometer sensor and	Collect accelerometer sensor data	Principal Component Analysis (PCA) for dimensionality	MLP classifier performs best with above 99% for precision, recall and F1-score	Utilize one accelerometer sensor that can change orientation and position of device at all time	(Sukor et al., 2018)

machine learning classifiers	embedded in a smartphone	reduction. Decision tree, SVM and MLP classifier for classification process			
A Comparative Study on the Influence of Undersampling and Oversampling Techniques for the Classification of Physical	Accelerometer data collected from wearable devices to classify physical activities based on ankle, hip and wrist movement	Random Forest and AdaBoost for ML, Random Undersampling (RUS) and SMOTE for imbalance data treatment	Overall classification performance is higher than 95% using both RUS and SMOTE approach. RUS is preferred due to low computational cost with similar result tendencies	Focuses on lower limb motion, thus classification accuracy is high on ankle movement but performances declined with hip and wrist movement data addition	(Jeong et al., 2022)

Activities Using an Imbalanced Accelerometer Dataset					
Activity Classifier: A Novel Approach Using Naive Bayes Classification	Movement data captured through Light Blue Bean device incorporated with a Bluetooth module and a tri-axial acceleration sensor	Naïve Bayes Classifier used for efficient classification with low input data	Naïve Bayes achieved accuracy of 90.67%	Low variability of activities limits generalizability	(Muneeswari et al., 2020)

Based on these previous studies, there are several essential research gaps revealed in handling accelerometer data for a better RLS prediction via machine learning. Some previous study focuses on using electrodermal activity (EDA), blood volume pulse (BVP), heart rate, skin conductance, and electroencephalogram (EEG) for anxiety or related psychological disorder prediction. Utilizing accelerometer data may provide a new potential discovery and could benefit from leveraging such multimodal signals. In addition, reliance of self-reported questionnaire introduces biasness towards prediction, hence limiting the accuracy and precision of prediction. Thus, such implementation will be avoided with the prediction will solely rely on the accelerometer data collected. Moreover, the occurrence of dataset imbalance is high in real time data collection, subsequently affecting model's performance scores. Studies have also indicated the lack of activity variation in prediction of daily activities which hinders the robustness of the model and does not provide proper conclusion. Thus, RLS prediction will incorporate data of participants exhibiting different activities that may trigger RLS phenomena.

Previous researches have highlighted potential suggestions for best approaches towards gaining real world implementation insights and creating a more accurate and robust model. Utilization of Fast-Fourier Transformer (FFT) in extraction of periodic features appears to be relevant in gaining insights from accelerometer data. This approach could be leverage for RLS classification where periodic leg motions marks as an essential aspect of the prediction. Sampling approaches such as Synthetic Minority Oversampling Technique (SMOTE) and Random Undersampling (RUS) can be considered to tackle imbalanced data issue, succeedingly enhance the reliability of model created. Machine Learning models such as K-nearest Neighbor (KNN), Naïve Bayes (NB), Support Vector Machine (SVM), as well as deep learning models such as Multilayer Perceptron (MLP) were heavily favoured in accelerometer data classification process. Performance scores such as accuracy, precision, recall and F1-score

were consistently used across various studies in performance evaluation, subsequently making it a suitable prediction model evaluation. Overall, featured research gaps and highlighted suggestions from the past studies can contribute on development of a more holistic and accurate prediction of anxiety through RLS detection.

Chapter 3: Research Methodology

Constructing methodology is an essential component of research as it serves as a breakdown of guidelines to provide a solution for research's problem statements, subsequently reaching the objective of research.

This research revolves around predicting anxiety disorder using accelerometry data from RLS and non-RLS patients. OSEMN framework is implemented for this project, which involves Obtain, Scrub, Explore, Model and Interpret due to its suitability for real-time analysis (Kumari et al., 2020). Python will be utilized as the main programming language within the process. Further details will be discussed within the following sections.

3.1 Research Design

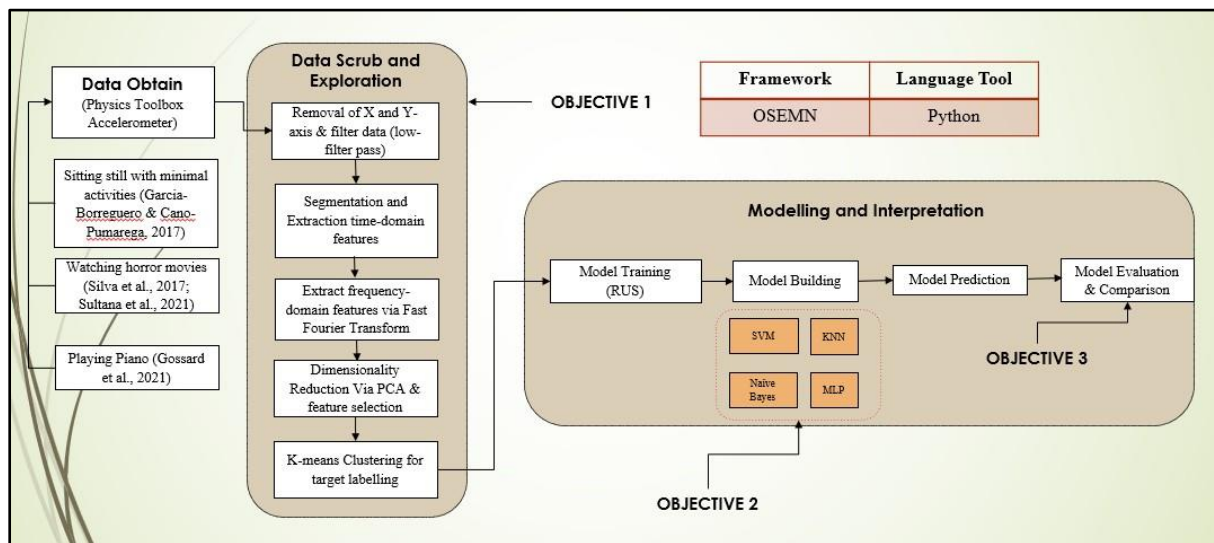


Figure 1: Research Design of Project

3.2 Data Obtain

The leg accelerometry data of individual with and without restless leg syndrome will be obtained via smartphone sensors, utilizing an app called advanced physical toolbox. Such visual programming was designed to established connection with sensors and retrieve the

accelerometry data. In conjunction with that, Kinematics g-force sensors, a type of gyroscope sensor, will be utilized to generate data. The smartphone consists of sensors with the ability to capture movement data in cartesian coordinate manner (XYZ format), therefore acting as the point of data gathering by capturing the frequency of data in graph format. This is essential as it determines the difference of leg movements between an individual with and without restless leg syndrome.

Previous study has indicated the tendency of periodical leg movements to occur during sedentary behaviour such as sitting still in RLS individual is more significant compared to normal individual and increases with longer period of time (Garcia-Borreguero & Cano-Pumarega, 2017). Furthermore, RLS can also be induced under the influence of emotional arousal triggered by fear such as watching horror movies (Silva et al., 2017; Sultana et al., 2021). Another study have also provide potential correlation of low dopamine activities as an RLS trigger point (Gossard et al., 2021). Thus, data capturing will be focused specifically during behaviours such as:

1. Sedentary behaviour while doing minimal activities such as working on laptop, scrolling on social media with individual's smartphone
2. Watching horror movies
3. Playing piano (an example of low dopamine activities)

The captured data is unstructured, comprising XYZ axis movement data from a smartphone's accelerometer. The targeted variable for this project is particularly on the Z axis due to its relevance in detecting leg motions as suggested from previous research (Reimers et al., 2021). Such motion is crucial to identify features related to restless leg syndrome including related

patterns, intensities or frequencies of movement. Thus, movement data of X and Y-axis will be filtered out to reduce complexity of subsequent processes.

3.3 Data Scrub

This section begins with the preparation and cleaning of data for further exploration. The initial process covers on identifying outliers, handling missing values and data normalization. It is essential to uncover outliers before handling missing values as result may deviate significantly during modelling process via machine learning. The data will be cleaned via removing redundant invaluable data for analysis. Such redundancy includes non-stationary smartphone movements, signals indicating intensive activities such as running or stomping, as well as high frequency data. Missing values are also handled by utilizing methods such as interpolation to replace missing data or discarding incomplete data segments.

Previous study suggested utilizing a low-pass filter is required to remove high-frequency noise from the accelerometer data, ensuring that the relevant low-frequency components are retained for analysis (Fridolfsson et al., 2019). A low-pass filter is a signal processing tool that filter signals by only allowing those with a frequency lower than its threshold to pass through while attenuating signals with higher ones. By applying a low-pass filter, this process aids in smoothening the data frequency and highlight the fundamental signals related to induced RLS leg motion. Considering the project revolves around data collection of sedentary and light activities, the optimal cutoff frequency suggested would be around 10Hz.

3.3.1 Low Pass Filtering

Low-pass filtering is a fundamental tool utilized in signal processing to allow relevant low frequency components to pass through while restricting high frequencies components.

Application of this process includes noise reduction and biomedical signal analysis in context of this project. This technique permits frequencies below a specified frequency value, known as the cutoff frequency, to pass through while hindering frequencies higher than the value. Considering a high potential of noise presence in collected accelerometer data, this process can increase the accuracy of model by removing potential noise as well as smoothen signals of the data, thus providing a higher data clarity and usability. A fourth order of butterworth low pass filter was suggested for this research, with the formula presented as below:

$$H(s) = \frac{1}{(s^2 + \sqrt{2}s + 1)^2}$$

Where $s = jw/w_c$ in a normalized Laplace variable. In python, the specified low pass filter can be implemented via *butter_lowpass* class imported from *scipy* library.

3.4 Data Exploration

The modelling process can be simplified and exhibit higher efficiency in prioritizing these features, which potentially can be beneficial for real-time application where efficiency and processing speed are both very important. In order to acquire a deeper understanding of the pre-processed data, an exploratory data analysis (EDA) will be conducted before implementing feature extraction.

3.4.1 Feature Extraction

Feature extraction is a crucial step in model development of RLS prediction as it bridges the gap in obtaining meaningful insights from raw data. RLS provides physiological anomalies

and periodic leg movements as key elements in extracting essential features for a reliable detection model.

The extraction focuses on two different domains, namely time domain and frequency domain. While time domain features can be extracted directly after filtering process, frequency domain features require implementation of Fast-Fourier Transformation (FFT) approach in altering the domain to frequency based. Suggestion from previous studies provided a total of 30 distinct and relevant features in isolating patterns and trends associated with RLS (Figo et al., 2010; Palmerini et al., 2011)

Discrete Transformation Function & Fast Fourier Transformer

The Discrete Fourier Transform (DFT) involves implementation of mathematical techniques in altering the time domain data to frequency domain via transforming discrete signals . DFT shows the phase and amplitude of distinct frequency in the signal by segmenting a finite sequence of N complex or real integers into a sum of sinusoidal components . The formula on DFT of a discrete sequence can be summarized as follows:

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}, \quad k = 0, 1, \dots, N-1$$

Where $X[k]$, $x[n]$ and $e^{-j\frac{2\pi}{N}kn}$ represents frequency-domain representation, time domain's input signal and basis of complex exponential function respectively (Cooley & Tukey, 1965).

Fast Fourier Transform (FFT) was innovated as an alternative and optimized algorithm to curb the computational inefficiency of the DFT due to $O(N^2)$ time complexity for larger dataset. The FFT leverages the symmetry and periodicity properties of the complex exponential functions used in the DFT by computing the DFT in $O(N\log N)$ time instead., the Cooley-

Tukey algorithm is the most commonly implemented across all FFT variations, which involves recursive division of FFT computation to a smaller subproblem. The formula of Cooley-Turkey algorithm is as below:

$$X[k] = \sum_{n=0}^{\frac{N}{2}-1} x[2n]e^{-j\frac{2\pi}{N}k(2n)} + \sum_{n=0}^{\frac{N}{2}-1} x[2n+1]e^{-j\frac{2\pi}{N}k(2n+1)}$$

In python, FFT can be utilized using *fft* class imported from *numpy* library. This allows FFT to become an essential tool in feature engineering by efficiently bridging the time and frequency domains of data.

Statistical Features

Statistical features focus on providing general distribution of signals to indicate any deviations & consistency in data collected. Table 2 indicates the 10 proposed statistical features and its description:

Table 2: Description of statistical features

Feature	Description
Mean	Average amplitude value of each segment
Median	Median amplitude value of each segment
Maximum	Maximum amplitude value of each segment
Minimum	Minimum amplitude value of each segment
Range	The difference value of maximum and minimum value of each segment
Root Mean Square (RMS)	Quadratic mean of each segment values
Skewness	Asymmetry measurement of each segment distribution
Kurtosis	Measure of “tailedness” of each segment distribution

Standard Deviation	Measure of variability of each segment
Variance	Squared deviation of each segment

Motion Features

Motion features revolve on highlighting movement irregularities via analysing dynamics of motion in obtaining any abrupt, repetitive or jerky movements. These types of features are time-sensitive as it captures behavioural changes over time. Table below describes the 11 suggested motion features and its function:

Table 3: Description of motion features

Feature	Description
Energy	Cumulative squared values of each segment
Zero Crossings	Total points where a signal passes through a specific value corresponding to half of the signal range
Angular Velocity	Average rate of change in each segment
Signal Magnitude Area	Sum of area encompassed by magnitude of signal in each segment
Signal Vector Magnitude	Square root of cumulative squared values
Jerk Index	Rate of change of acceleration, also known as index of smoothness
Normalized Jerk Index	Jerk index divided by squared of sway path
Sway Path	Total centre of mass (CoM) trajectory length (mm)
Sway Area	Area under the curve of CoM displacement per second (mm ² /s)
Mean Velocity	Median of absolute value obtained through centre of mass displacement in time series
Maximum Peak Distance	Largest distance between consecutive signal peaks

Fourier-transformed Features

Fourier-transformed or frequency-based features concentrate on measuring complexity and variability in frequency domain, as well as highlighting dominant frequencies and power in movement signals. These aspects are essential in identifying frequency bands associated with RLS movements. Table below indicates the 9 proposed fourier-transformed features and its function:

Table 4: Description of Fourier-transformed features

Feature	Description
Dominant Frequency	Frequency with highest magnitude (Hz) from each segment
Mean Power Spectral Density	Average power across frequencies in each segment
Maximum Power Spectral Density	Highest frequency power in each segment
Coefficient Sum	Sum of absolute FFT coefficients
Power Spectral Density Entropy	A measure of disorder in frequency distribution in a segment
Frequency Dispersion	A measurement of variation of PSD frequency content (unitless)
Centroidal Frequency	The value of frequency with highest concentration in spectrum (Hz)
F50	Frequency of 50% cumulative power for each segment
F95	Frequency of 95% cumulative power for each segment

Overall, combining these 3 feature classes can provide a comprehensive analysis on movement trends to efficiently differentiate RLS and non-RLS patterns.

3.4.2 Feature Selection

Feature Selection, also known as variable selection, is a crucial step in the modeling process. It involves identifying and selecting the most relevant variables that contribute significantly to the prediction and output of interest. The main goal of feature selection is to identify related variables and improve model's performance by reducing overfitting, improving accuracy, and reducing training time.

In the context of this study, feature selection is particularly important due to the multivariate nature of data, with multiple potential predictors available from the streamflow data, it is crucial to identify what types of variables needed and has significant impact on the output, which in this case of motion reading using statistical features. Based on a research, feature selection can also provide a lower processing time consumption while maintaining the model's performances particularly on human activity recognition studies (Wang et al., 2016).

By carefully referring literature, exposing the purposed features from the data that are important for our model. Previous study have indicated the implementation of principal component analysis (PCA) in classifying accelerometer data.

3.4.3 K- Means Clustering

K-Means Clustering is a common unsupervised machine learning algorithm which is mainly utilized for clustering task in search patterns and group data points with high similarity. The fundamental principle of K-Means involves partitioning a dataset into desired number of clusters known as k . Iteratively, the operation of algorithm starts by initialization of k centroids randomly or implementing advanced initialization methods. Subsequently, distance metrics such as Euclidean distance is used to allocate each data points to the nearest initialized centroid.

The centroids are then recalculated as the data points' average allocated to each cluster. The process continuously repeated until convergence is indicated via centroid stabilization or zero change in assignments. The objective is to maximize the separation between clusters while reducing the within-cluster sum of squares (WCSS) in determining the data points are feasibly similar within each cluster.

A well-known metric in assessment of clustering outcomes' quality for clustering algorithm, including K-means, would be silhouette score. It determines the quality of allocated data points within the assigned cluster by taking into account inter-cluster separation and intra-cluster cohesion respectively. For a given data point i , below are the respective calculation of silhouette score:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))'}$$

Where

$a(i)$ is the mean distance between i and every other data points in its respective cluster, representing cohesion

$b(i)$ is lowest average distance between i and data points in any different cluster, representing separation

Ranges of exhibited silhouette score are from -1 to 1, with higher values suggests a higher quality cluster and better defined while lower values of the range indicating poor separation and allocation as well as potential misclassification. This scoring matrix is extremely beneficial in K-means clustering for determination of number of clusters (k) as well as quality of cluster model in terms of separation between other clusters and cohesion within own cluster.

3.5 Modelling

This section focuses on data partitioning and description of machine learning methods, including on RLS classification process.

3.5.1 Data Partitioning

Data partitioning is a crucial phase in the modelling process which entails splitting given dataset into distinct subsets in ensuring model development robustness and generalizability. The data is usually divided into training, test, and in some cases validation set. The training set fed into the model in order to identify patterns in data. The validation set is utilized for modelling in prevention of overfitting by training data by assessing performance and fine-tune hyperparameters. Subsequently, remaining data is reserved for the testing set and categorized as unseen dataset to evaluate the model's final performance and generalization ability. Random splitting, stratified sampling, or time-based splitting are well-known techniques implemented according to the data nature and modelling objectives. In addition, appropriate data partitioning contributes to less biased model assessment and aids in developing predictive models with relatively high performance on new, unseen data (Géron, 2019). In python, data partitioning can be implemented using *train_test_split* class imported from *sklearn* library.

3.5.2 Naïve Bayes (NB)

Naive Bayes is a basic yet greatly effective supervised machine learning algorithm that utilizes the probability concept of Bayes' Theorem and broadly implemented for classification tasks in various domains including spam detection and medical diagnosis. This algorithm creates a “naive” assumption which features are conditionally independent given the class label, thus providing robust results frequently in practice. Bayes' Theorem offers a probabilistic foundation for classification via determination of posterior probability of a class given the observed features. Naive Bayes boosts the precise classification probability via assigning a new data point to the class with the highest posterior probability. There are different types of Naïve Bayes implementation based on the data type with Gaussian Naïve Bayes and Multinomial Naïve Bayes are utilized for continuous and discrete, count-based data respectively. It also exhibits great computational efficiency, is relatively well in terms of scalability, and produces high performance in situations where independence assumption are more likely true (Minsky, 1961; Russell & Norvig, 2003). This study focuses on utilizing Gaussian Naïve Bayes for classification process via *GaussianNB* model imported from *sklearn.naive_bayes* class in python.

3.5.3 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a powerful and versatile supervised machine learning algorithm, primarily used for classification and regression tasks. SVM in its modern form has been broadly practiced in multidisciplinary fields from science perspective of bioinformatics to social perspective such as natural language processing. The main principle

of SVM involves determining most suitable decision boundary known as separating hyperplane, that maximizes the margin between different classes in the training data. In terms of two-dimensional concept, such hyperplane can be idealized as a line separating a plane into two sections, with each class located on either side. The objective is to locate the line with the largest distance to the nearest points in the training data, known as support vectors. This notion ensures that the classifier not only distinguishes the classes but does so in the most clear-cut way possible, contributing to the model's resilience. (Cortes & Vapnik, 1995). The SVM classifier can be utilized by importing SVC model from *sklearn.svm* library in python.

3.5.4 K-nearest neighbor

K-Nearest Neighbors (KNN) is a supervised machine learning algorithm that utilizes instance-based learning concept. It is commonly used technique across various domains including medical diagnosis for either classification or regression tasks. KNN's fundamental basics centres on identifying similarity between data points in making predictions. In depth, the the k nearest data points, or also known as 'Neighbors', is identified to a given query point from determination via selected distance metric, namely Euclidean, Manhattan, or Minkowski distance. In context of classification, the allocation of query point is based on the query point is allocated to the highest class frequency among its neighbors. This allows KNN to achieve flexibility and intuitional by having a natural adaptation to complex decision boundaries (Cover & Hart, 1967; Kramer & Kramer, 2013).

3.5.6 Multi-layer Perceptron

Multilayer Perceptron (MLP) is a powerful and versatile type of artificial neural network, widely utilized for supervised learning tasks such as classification, regression, and even time-series forecasting. The fundamental basis of an MLP consists of an input layer, at least one hidden layer, and an output layer, with interconnected nodes known as neurons are present in each layer. MLP has the ability to identify intricate patterns and relationships due to these neurons' utilization of weighted connections, biases, and non-linear activation functions in processing information. The universal approximation theorem illustrates the core idea of MLP, which is that any continuous function may be approximated using its layer's structure. MLP data training revolves around adjusting its weights and biases via optimization algorithms like stochastic gradient descent, in compliance with the error signal calculated through backpropagation. As a result, minimization of dissimilarity between predicted and actual outcome can be achieved via this iterative process. Such process allows the ability to model intricate and non-linear relationships, potentially becoming the fundamental tool in modern machine learning (Popescu et al., 2009; Rumelhart et al., 1986)

3.6 Evaluation and Interpretation

There are 4 evaluation metrics considered for this evaluation which are:

1. Accuracy:

Accuracy measures the proportion of correct predictions, which is true positives and true negatives, out of all predictions made. It is a good measure when the class distributions are balanced. The formula is as below:

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN}$$

2. Precision:

Precision calculates the fraction of correct positive identification. It indicates the ability of a model to identify relevant instances correctly. This is important, particularly in cases where exhibiting a higher false positive is more detrimental than false negative. The formula is as below:

$$\text{Precision} = \frac{TP}{TP+FP}$$

3. Recall:

Recall indicates the actual positives proportion that were determined correctly. It assesses the model's ability to pinpoint each relevant instance. This is significant if missing a positive instance (false negative) is proven to be more costly than incorrect labelling of negative instance as positive (false positive). The formula is as below:

$$\text{Recall} = \frac{TP}{TP+FN}$$

4. F1-score

The F1-score indicates a balance between the two metrics of precision and recall. This measurement is significant when both false positives and false negatives are equally detrimental. The formula is as below:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Chapter 4: Results and Discussion

This section mainly discusses on result of exploratory data analysis conducted, process of model builds and its result on 3 different activities based on evaluation metrics proposed in methodology section. 3 different activities included are sitting still with minimal movement activities, watching horror movies and playing piano, which were characterized as activity 1, activity 2 and activity 3 respectively. A sliding window size 128 with a step size 64 was utilized over a segmented time series data. A fourth order of Butterworth low pass filter was implemented for signal processing to reduce noise and smoothen signals. A frequency domain data was engineered through Fast-Fourier transformation process from the filtered time domain data. A total of 30 features were extracted consisting of statistical, motion and frequency-based features from time and frequency domain of filtered and transformed data. Activity 1 was utilized as a baseline considering its subjected activity is the most common practice in triggering RLS phenomena.

4.1 Exploratory Data Analysis

Univariate analysis of 30 extracted features from activity 1 were conducted via histogram charts and box plot. Such analysis is essential to determine the context of feature extraction for RLS prediction behaviour. This analysis aids in determining the variation, central tendencies and characteristic distribution of data to detect any underlying physiological anomalies in most cases. It also indicates potential redundancies and noises present in data, subsequently provides opportunity for data normalization and transformation if required. Figure 2 and 3 show histogram charts and box plot series of the 30 extracted features:

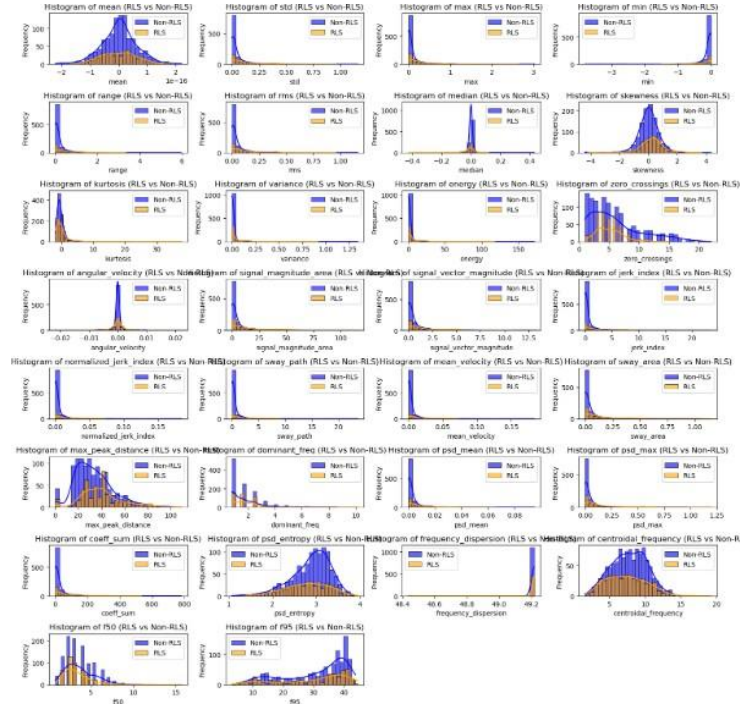


Figure 2: Histogram charts of extracted features of RLS and non-RLS data

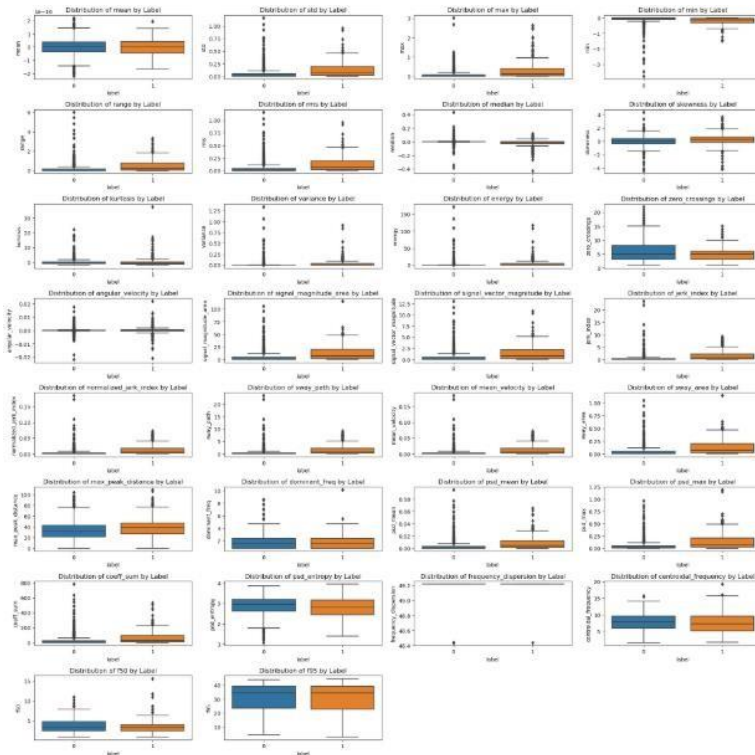


Figure 3: Whisker boxplot of extracted features of both RLS and non-RLS data

While features such as mean, median, skewness and angular velocity indicates a normally distributed data from the histogram charts, most features experienced a skewed data distribution. Furthermore, the box plot diagram indicates the presence of outliers in certain features. However, some data points may be due to the presence of skewed distribution rather than extreme values. Nevertheless, this highlights the requirement of data transformation and normalization for subsequent process for a more reliable modelling prediction. Techniques such as z-score normalization and box-cox transformation were selected as it aids in patterns anticipation without biasness towards larger magnitude features as well as making data more symmetric as subjected from past research (Weiss & Lockhart, 2012). The result after data transformation is depicted in a box plot series as diagram below:

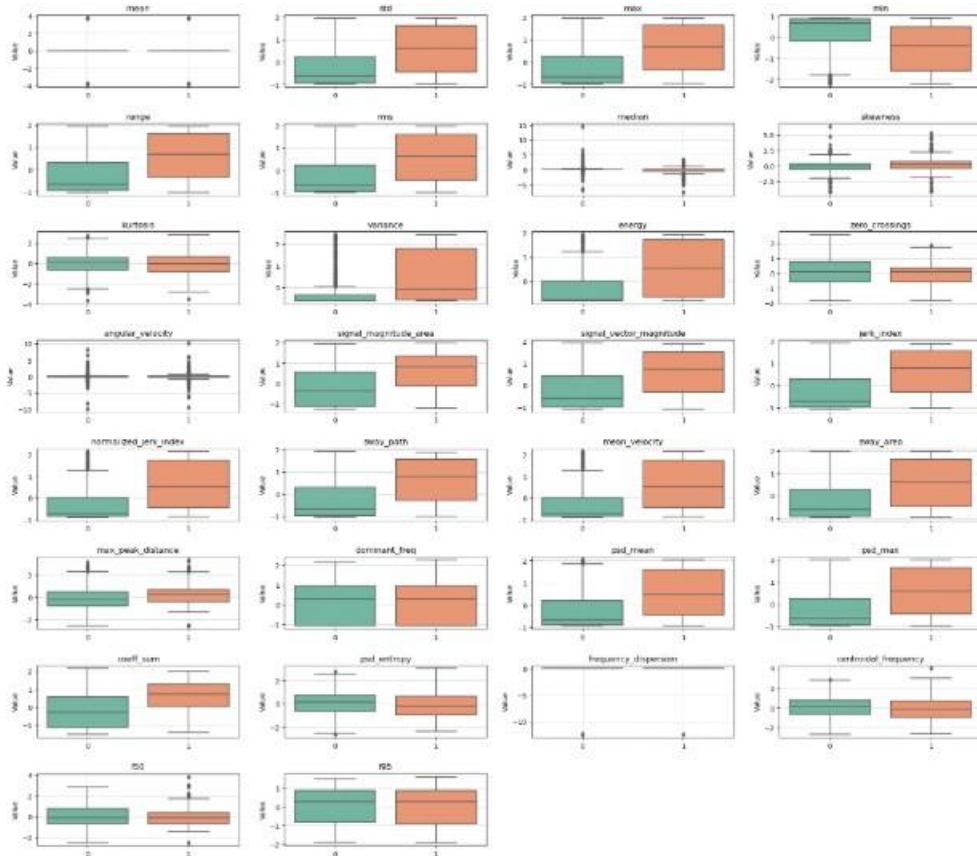


Figure 4: Whisker boxplot of extracted features (after data transformation)

Subsequently, feature selection was conducted via principal component analysis (PCA) and feature importance-based selection with random forest algorithm to determine the optimum number of features for model building as well as identifying the most impactful feature in terms of contribution. Results from Figure 5 highlighted potential of reduction dimensionality to 10 features to retain 95% of variance for a reliable model. Furthermore, statistical features such as min, max, median, skewness and kurtosis, as well as 5 motion features of jerk index, normalized jerk index, sway path, zero crossings and mean velocity were selected based on feature importance score in Table 5. The top 10 feature importance scores range from 3.87% to 6.71%, indicating a comparable contribution among features towards the predictive model. This low variation of importance score can be advantageous as there are no overwhelmingly dominant features presence for models' prediction, potentially improving generalization and robustness.

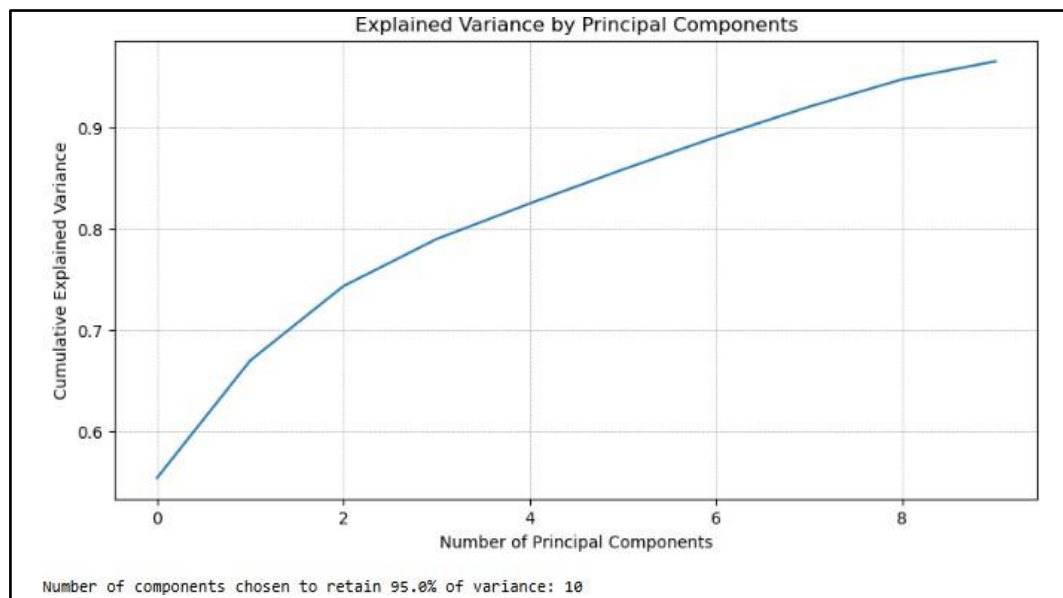


Figure 5: PCA plot

Table 5: Feature importance based-selection score

Top Important Features	Feature Important Score
Jerk Index	6.71%
Max	6.12%
Sway Path	5.80%
Normalized Jerk Index	5.50%
Mean Velocity	5.14%
Zero Crossings	4.50%
Median	4.06%
Min	4.04%
Kurtosis	3.93%
Skewness	3.87%

K-means clustering was developed for 3 activities based on the selected features to classify data into labels of RLS and non-RLS. Results on data distribution and silhouette score of clusters of 3 activities were shown in Table 6. Activity 1 data distribution consists of 66.69% of non-RLS and 33.31% of RLS with a clustering silhouette score of 0.416. The percent difference of activity 1 is the highest among 3 activities at 33.38%, indicating the clearest distinction between 2 groups. Activity 2 has the least data imbalanced with 15.66% percent difference in data distribution, and a slightly lower silhouette score of 0.404. This highlights the selected features from activity 1 can separate clusters of activity 2 almost as efficient as activity 1, thus having clusters that can generalized well. Activity 3 possess the lowest silhouette score of 0.357, indicating less distinct of cluster separation compared to other 2 activities. All 3 activities suffers from data imbalanced issues as expected based on previous study (Abd Al-Alim et al., 2024). Thus, data balancing will be issued in the modelling process.

Table 6: Data distribution and silhouette score for clusters of different activities

Activities	Data Distribution					Silhouette Score
	Instances		Percentage (%)			
	Cluster 1 (Non-RLS)	Cluster 2 (RLS)	Cluster 1 (Non-RLS)	Cluster 2 (RLS)	Percent Difference	
Activity 1 (Sitting Still)	1039	519	66.69%	33.31%	33.38%	0.416
Activity 2 (Watching horror movies)	1271	927	57.83%	42.17%	15.66%	0.404
Activity 3 (Playing Piano)	736	486	60.23%	39.77%	20.46%	0.357

4.2 Modelling

Previous research suggest employing random undersampling technique (RUS) as an effective measure for addressing data imbalance (Wongvorachan et al., 2023). The related study indicated their preference over SMOTE (Synthetic Minority Oversampling Technique), an oversampling technique for data imbalance treatment, when classifying activities via accelerometer data due to low biasness, computation efficiency and comparable results with the oversampling technique. This technique was implemented using the RandomUnderSampler class from imblearn library, allowing dataset balancing by randomly removing the data points of majority class to match the volume of minority classes. Thus, final data instances are as shown in Table 7. Consequently, the dataset of each activity was split into train and test set at a percentage ratio of 70% and 30% respectively.

Table 7: Number of clusters' instances before and after RUS

Activities	Number of Instances (Pre RUS)		Number of Instances (Post RUS)	
	Cluster 1 (Non-RLS)	Cluster 2 (RLS)	Cluster 1 (Non-RLS)	Cluster 2 (RLS)
Activity 1	1039	519	519	519
Activity 2	1271	927	927	927
Activity 3	736	486	486	486

4.3 Performance Result

4.3.1 Activity 1: Sitting still

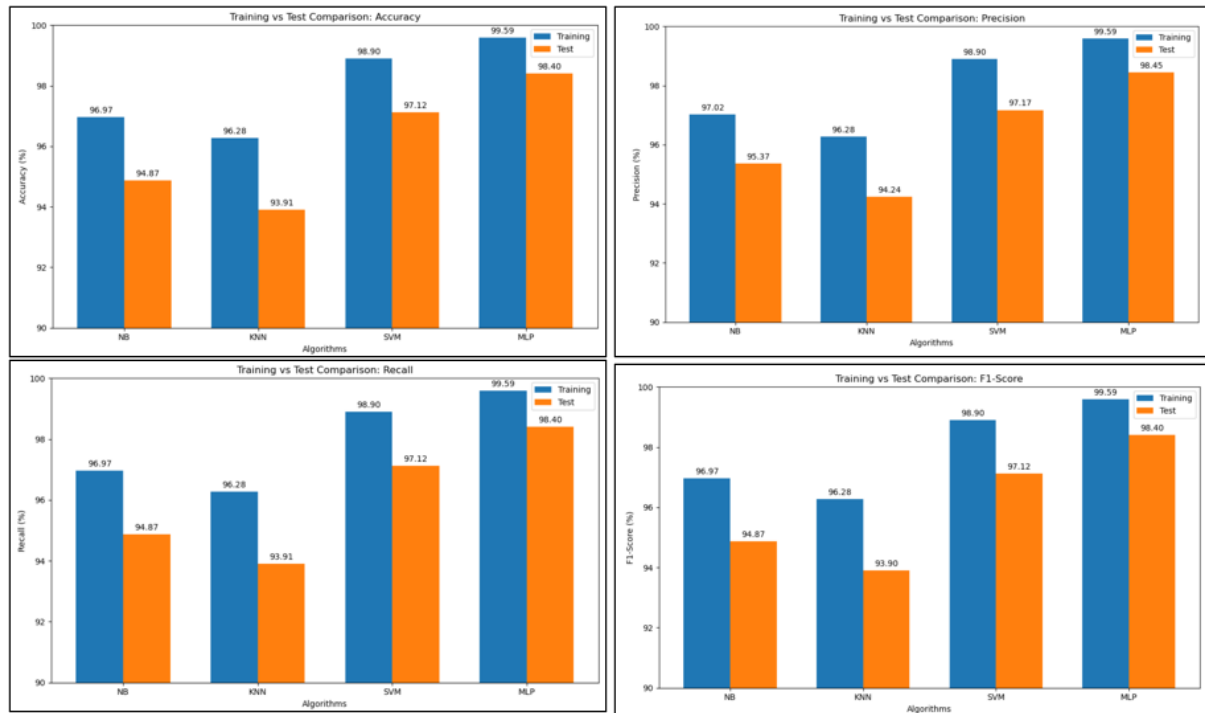


Figure 6: Evaluation metrics histogram plot for training and test set of activity 1

Based on Figure 6, Multilayer Perceptron (MLP) model exhibited highest performance score in accuracy, precision, recall and F1-score with value of 99.59% each in training set. Similarly, MLP also produce best performance in all aspects of test set, with value ranging from 98.40% to 98.45%. In contrast, K-Nearest Neighbor (KNN) model exhibited lowest score in training set compared to others, with each performance score valued at 96.28% for each

metrics. The testing set performance scores of KNN also significantly drop to range of 93.90% to 94.24%. Despite having better performance than KNN model, Naïve Bayes (NB) model displays similar patterns with performance scores, having a much higher performance score in training set ranging from 96.97% to 97.02%, to a quite significant drop in test set with the scores ranging from 94.87% to 95.37%. SVM performs relatively better with score of 98.90% for each performance metrics in training set followed by a score range of 97.12% to 97.17% for test set.

4.3.2 Activity 2: Watching horror movies

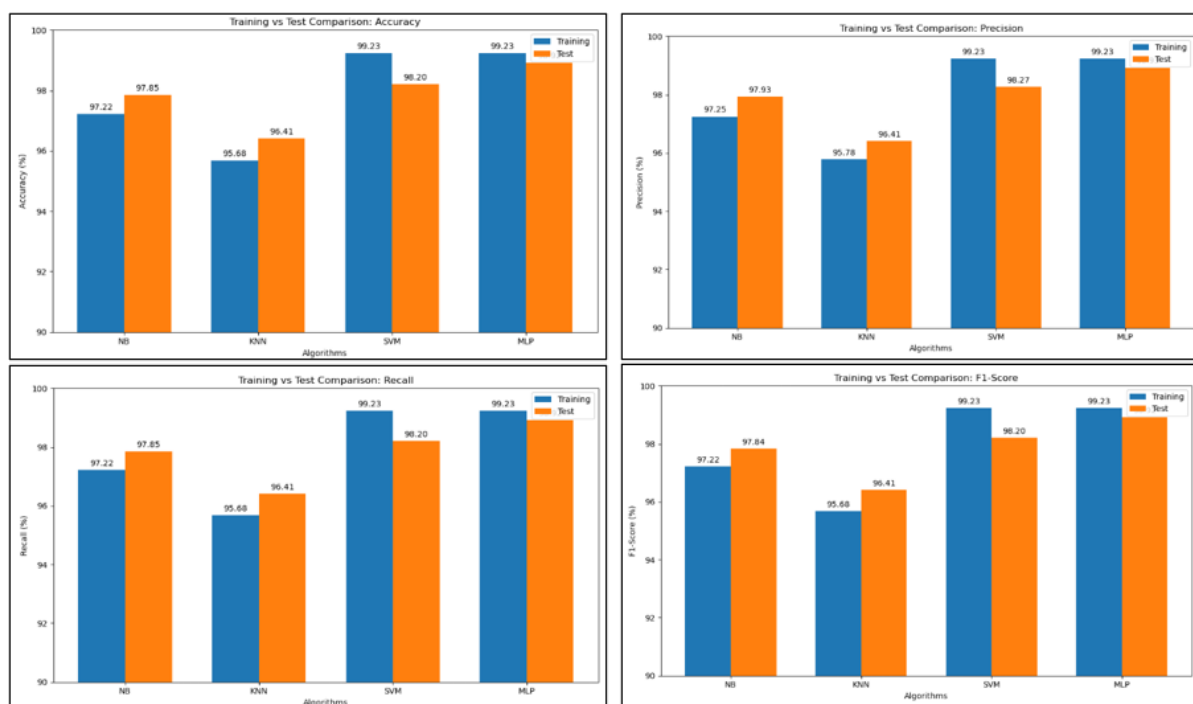


Figure 7: Evaluation metrics histogram plot for training and test set of activity 2

The result in Figure 7 displays a recognizable pattern from activity 1 in which MLP presents the highest performance scores in all metrics for both test and training set. SVM performs as well as MLP in training set, with both scoring 99.23% for all performance metrics, but a slightly lower score performances in test set with 98.20% for accuracy, recall and F1-score and 98.27%

for precision. However, NB and KNN manifests similar trend of test set performance scores are higher than training set. This could highlight the presence of underfitting issue in both models resulting from models' simplicity and inability to capture complex pattern and relationships within the data.

4.3.3 Activity 3: Playing Piano

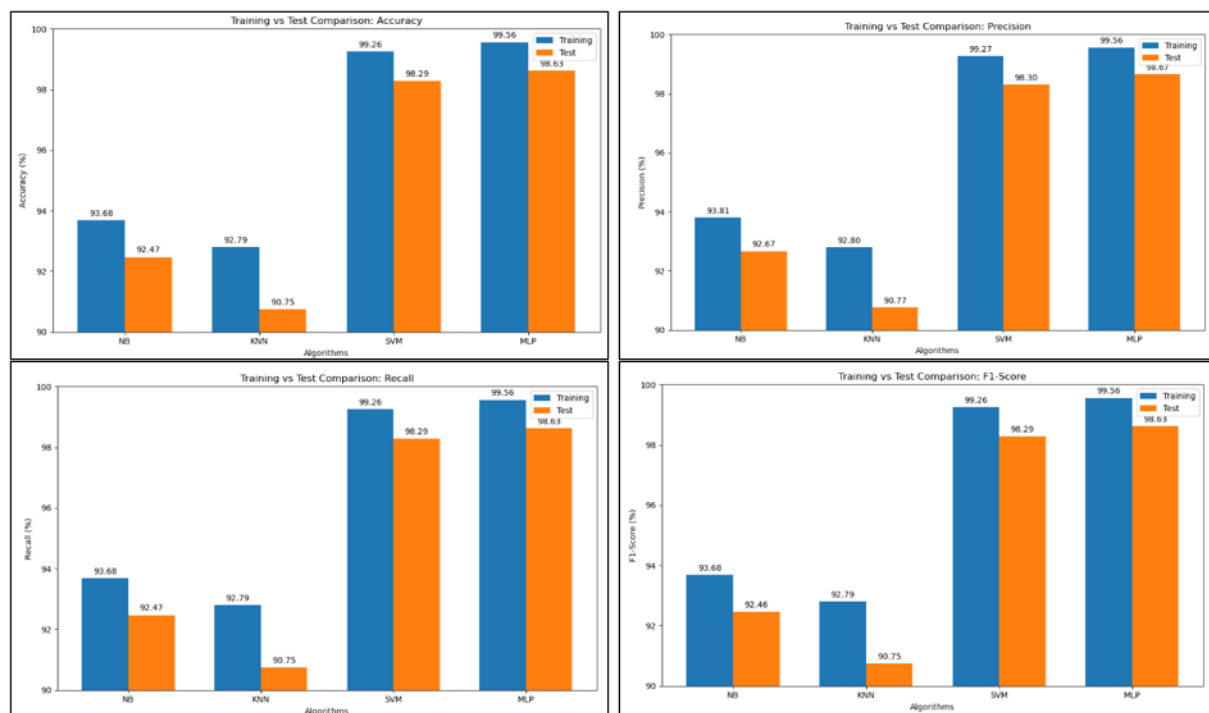


Figure 8: Evaluation metrics histogram plot for training and test set of activity 3

Figure 8 indicates similar scenario with MLP has the top performance scores in both training (99.56%) and test (98.63% - 98.67%) set. SVM performs relatively well with a range performance score of 99.26% to 99.27% for training set with a slightly lower performance score ranging from 98.29% to 98.30% for test. Albeit, there is a significant drop of performance scores in both NB and KNN model. NB model exhibits range performance scores of 93.68% to 93.81% for training set, which further drops between range of 92.46% to 92.67% for test set. Furthermore, KNN exhibits lowest performance scores in training set with a range of 92.79%

to 92.80%. Similarly, the performance scores drastically drop to a range between 90.75% to 90.77% in test set.

Overall, MLP is the best model due to its ability to generalize well in RLS detection with testing data across 3 different activities and consistently scores above 98% across all metrics. SVM generally performs well, maintaining a high-performance scores across 3 different activities with only a slight decline in testing performances. However, both NB and KNN highlights poor performance with variation of scores across 3 activities as well as potential underfitting issue in RLS detection particularly in activity 2. Moreover, KNN displays lowest performance scores for all 3 activities. This might be due to activity variability and dynamic movement patterns opposes challenge towards these models in terms of capturing non-linear and complex trends as these models rely on simpler decision boundaries and distance metrics. Hyperparameter optimization and regularization would be a requirement for such models to exhibit a better performance.

Chapter 5: Conclusion

5.1 Achievements

The result of this study have addressed the following research questions (RQ) outlined in section 1.3

RQ1: How can accelerometer data captured from leg movements be utilized to differentiate between states of anxiety and non-anxiety?

States of anxiety and non-anxiety can be differentiated through capturing leg data movements of RLS diagnosed participants along with non-RLS diagnosed participants. The captured data was filtered, feature extracted and labelled to two classes consisting of RLS and non-RLS during clustering process.

RQ2: How can machine learning techniques aid in detection of RLS through accelerometer data captured?

Incorporating machine learning techniques can enhance the detection of RLS by leveraging the accelerometer data labelled from clustering process. While clustering is implemented for initial labelling, machine learning techniques enables development of models that can learn complex and non-linear relationships between extracted features and RLS. This ability allows the model to classify RLS and non-RLS even from an unseen data.

RQ3: How does the inclusion of diverse activity sources in accelerometer data impact the accuracy of machine learning models in predicting restless legs syndrome (RLS)?

The inclusion of diverse activity data significantly impacts the machine learning models' accuracy by increasing the generalizability and complexity of such models. Different activities introduces variations in movement patterns as well as poses challenge in distinguishing RLS movements with activity-induced noise. Overall, incorporation of different activities may

provide enhancement in terms of robustness and generalizability towards model for a real-time application.

In addition, the research objectives (RO) requirement were met as follows:

RO1: Data of leg movements were gathered from 3 RLS and 3 non-RLS participants respectively via a mobile phone app called Advanced Physical Toolbox. The data was filtered through a fourth-order butterworth low pass filter, feature extracted, feature engineered, normalized, clustered and labelled with RLS and non-RLS throughout the whole process

RO2: 4 machine learning models were developed based on suggestion from previous study, namely Naïve Bayes (NB), K-Nearest Neighbor (KNN), Support Vector Machine (SVM) classifier and Multilayer Perceptron (MLP) classifier. These models purpose were to classify the initially labelled data from clustering.

RO3: A comparison was conducted between models based on 4 key metrics, which are accuracy, precision, recall and F1-score, across 3 different activities. The result indicated that simpler models such as NB and KNN were not able to generalized well with potential underfitting issue and significant drop in performance scores on some of the activities. However, advanced model such as SVM and MLP excel in classifying RLS with a relatively high performance scores across those activities, particularly for MLP consistently provide performance scores above 98% across all metrics for each of these activities. This indicate the presence of intricate non-linear patterns and complex relationships which is more suitable for utilization of advanced models.

In summary, this study not only provides an insight on RLS classification and prediction in aiding the detection of anxiety, but also opens a new path in developing a more reliable and favourable model for early detection of anxiety in prevention of subjected condition to worsen

5.2 Research Limitations

Through out the research process of utilizing accelerometer data in RLS classification for anxiety detection, there are several limitations encountered:

Demographic and data size limitation: Reliance of data from only 3 participants from each classes displays the challenges in providing a robust and generalization of prediction in RLS. Furthermore, the participants involved were particularly youth age people, thus providing a narrow representation in terms of movements variation. As a result, models developed may not accurately predict and generalized well with a broader population

Lack of feature correlation study: The absence of feature correlation study may induce multicollinearity in classification and prediction process, incurring a redundancy on the feature selected. This may result in reduction of model stability, accuracy and interpretability

Low silhouette score: Silhouette score for clustering model ranges from 0.357 to 0.416 across 3 activities, indicating a relatively poor score. Previous study indicated a minimum value of 0.7 in silhouette score to have better classification based on accelerometer data (Jones et al., 2019). This indicates that the cluster model formed are not well represented with presence of overlapping between clusters due to K-means assumption of spherical and equally sized clusters.

Reliance of single data source: While RLS have shown a potential relationship with anxiety due to similar mechanisms such as dopamine dysfunction and impact of sleep disturbances,

relying on single data source may oversimplify this complex relationship. There are other factors such as diseases (iron deficiency, diabetes and Parkinson diseases), medications (neuroleptics and antidepressants) and diet intakes (caffeinated drinks and alcohols) reported to trigger RLS (Mansur et al., 2025).

5.3 Future Works

Considering the findings and limitation of this study, several tasks can be implemented for future research, including:

- I. Expand participants demography: Future studies should prioritize expansion of demographic diversities of participants to improve generalizability and reliability of findings. Varying participants particularly from diverse age groups enables researchers in exploring variation of RLS in different life stages.
- II. Addition of feature correlation and regularization technique: Additional of feature correlation techniques such as Pearson's correlation coefficients or Variance Inflation Factor (VIF) may aid in handling multicollinearity issue. Incorporation of regularization technique such as Elastic Net could be beneficial as it is particularly designed to manage both multicollinearity and overfitting issues in feature selection process.
- III. Implement different clustering techniques: Considering the limitation of K-means clusters due to assumption of spherical clusters as well as imbalance data issues, utilizing alternative clustering techniques such as density-based clustering of application with noise (DBSCAN) or Gaussian Mixture Models (GMM) may deal with such constraint. DBSCAN possesses the ability to detect irregular cluster shapes and highly robust to noise while GMM accounts for different densities of cluster by

concerning on probabilistic assignments. These techniques allow a better cluster representation, leading to a more reliable model development.

- IV. Incorporate other physiological indicators: Recent study have developed a solid based detection relating to mental health issue by combining different physiological indicators that are non-invasively collected such as electrocardiography (ECG), skin temperature (ST), and skin conductance (Abd Al-Alim et al., 2024). Combining of any of these indicators or other related ones such as heart rate, blood pressure, oxygen saturation and respiratory rate with accelerometer data of leg movements collected may improve the accuracy of anxiety detection process for real-time detection.

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